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**B.Tech. Degree I Semester Regular and Supplementary /
II Semester Supplementary Examination November 2018**

CE/EE/ME/SE AS 15 - 1101 A & CS/EC/IT AS 15 - 1201 B CALCULUS

(2015 Scheme)

Time : 3 Hours

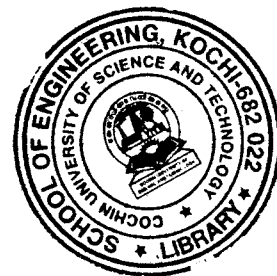
Maximum Marks : 60

PART A(Answer *ALL* questions)

(10 × 2 = 20)

I.

- (a) Solve $xy(1+xy^2)\frac{dy}{dx}=1$.
- (b) Solve $(5x^4+3x^2y^2-2xy^3)dx+(2x^3y-3x^2y^2-5y^4)dy=0$.
- (c) Solve $(D^3+1)y=0$.
- (d) Test the convergence of the series
 $1+\frac{1}{4^{\frac{2}{3}}}+\frac{1}{9^{\frac{2}{3}}}+\frac{1}{16^{\frac{2}{3}}}+\dots$
- (e) Test whether the series is absolutely convergent or conditionally convergent
 $1-\frac{1}{2^2}+\frac{1}{3^2}-\frac{1}{4^2}+\dots$
- (f) Expand $\cos x$ about $x=\pi/2$ using Taylor's series.
- (g) If $u=\tan^{-1}\frac{x^3+y^3}{x-y}$ prove that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\sin 2u$.
- (h) Show that $u=\frac{x+y}{x-y}$, $v=\frac{xy}{(x-y)^2}$ are functionally dependent.
- (i) Find the area of the curve $r=2a\cos\theta$.
- (j) Change the order of integration in the integral and sketch the region $\int_0^a \int_{x^2/a}^{2a-x} f(x,y)dx dy$.

**PART B**

(4 × 10 = 40)

- II. (a) Solve $\frac{d^2y}{dx^2}+y=e^{2x}+x\cos x$.
- (b) Solve $x^2\frac{d^2y}{dx^2}+2x\frac{dy}{dx}-20y=x^4$.

OR

- III. (a) Solve $\frac{dx}{dt}=2y-1$; $\frac{dy}{dt}=2x+1$
- (b) A voltage Ee^{-at} is applied at $t=0$ at a circuit containing inductance L and resistance R . Show that the current at any time t is $\frac{E}{R-aL}\left(e^{-at}-e^{-\frac{R}{L}t}\right)$ assuming that $i=0$ at $t=0$.

(P.T.O.)

- IV. (a) Determine the nature of series $\sum \left[\left(\frac{n+1}{n} \right)^{n+1} - \left(\frac{n+1}{n} \right) \right]^n$.
- (b) If $y = \cos(m \sin^{-1} x)$, prove that $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2 - m^2)y_n = 0$.

OR

- V. (a) Show that series $\sum_{n=0}^{\infty} \frac{(n!)^2 x^n}{(2n)!}$ is convergent for $x < 4$ and divergent for $x \geq 4$.
- (b) Expand $\sin x$ in powers of $(x - \pi/2)$. Hence find the value of $\sin 91^\circ$.

- VI (a) If u is a homogeneous function of degree n in x and y show that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u$.
- (b) Find the maximum and minimum values of $f(x, y) = x^3 + y^3 + 3xy$.

OR

- VII. (a) If $u = f(r, s, t)$ and $r = \frac{x}{y}$, $s = \frac{y}{z}$, $t = \frac{z}{x}$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$.
- (b) Find the shortest and longest distance from the point $(1, 2, -1)$ to the sphere $x^2 + y^2 + z^2 = 24$.

- VIII. (a) Find the volume of the solid generated by revolving $x^{2/3} + y^{2/3} = a^{2/3}$ about the x -axis.
- (b) Evaluate $\iint_R \frac{e^{-y}}{y} dx dy$ by choosing order of integration suitably given that, region 'R' is bounded by the lines $x = 0$, $x = y$, and $y = \infty$.

OR

- IX. (a) Find the area of the crescent bounded by $r = \sqrt{3}$ and $r = 2 \cos \theta$.
- (b) Evaluate $\int_0^1 \int_{y^2}^1 \int_0^{1-x} x dz dx dy$.
